

Avant balancement :

Métrique	XGBoost	AdaBoost	Logistic Regression	MLP	DNN
Accuracy	0.9402	0.5160	0.7543	0.8988	0.9148
Precision	0.9395	0.4643	0.7475	0.8957	0.9140
Recall	0.9402	0.5160	0.7543	0.8988	0.9148
F1-Score	0.9366	0.4476	0.7417	0.8895	0.9056
FAR / FPR	0.0032	0.0260	0.0132	0.0054	0.0046
ROC-AUC Score	0.9978	0.8940	0.9640	0.9935	0.9954
Matthews Corr. Coeff (MCC)	0.9331	0.4769	0.7260	0.8868	0.9048

Attack Type	XGBoost (Precision, Recall, F1-Score)	AdaBoost (Precision, Recall, F1-Score)	Logistic Regression (Precision, Recall, F1-Score)	MLP (Precision, Recall, F1-Score)	DNN (Precision, Recall, F1-Score)
Analysis	0.22, 0.31, 0.26	0.00, 0.00, 0.00	0.00, 0.00, 0.00	0.00, 0.00, 0.00	0.00, 0.00, 0.00
Backdoor	0.99, 0.96, 0.98	0.43, 0.00, 0.00	0.94, 0.95, 0.94	0.97, 0.95, 0.96	0.97, 0.96, 0.96
Benign	0.86, 0.94, 0.89	0.32, 0.54, 0.41	0.74, 0.51, 0.60	0.82, 0.90, 0.86	0.83, 0.92, 0.87
Bot	1.00, 1.00, 1.00	0.00, 0.00, 0.00	1.00, 1.00, 1.00	1.00, 1.00, 1.00	1.00, 1.00, 1.00
Brute Force	1.00, 0.98, 0.99	1.00, 0.65, 0.79	0.90, 0.97, 0.94	0.99, 0.98, 0.99	1.00, 0.98, 0.99
DDoS	0.99, 0.99, 0.99	0.99, 0.84, 0.91	0.94, 0.94, 0.94	0.95, 0.98, 0.97	0.98, 0.98, 0.98
DoS	0.96, 0.98, 0.97	0.92, 0.77, 0.84	0.91, 0.89, 0.90	0.94, 0.98, 0.96	0.94, 0.98, 0.96
Exploits	0.95, 0.96, 0.96	0.61, 0.01, 0.02	0.56, 0.30, 0.39	0.83, 0.92, 0.87	0.87, 0.90, 0.88
Fuzzers	0.96, 0.97, 0.97	0.00, 0.00, 0.00	0.63, 0.93, 0.75	0.83, 0.94, 0.88	0.88, 0.93, 0.90
Generic	0.83, 0.87, 0.85	0.00, 0.00, 0.00	0.87, 0.34, 0.49	0.93, 0.41, 0.57	0.58, 0.34, 0.43
Infiltration	0.83, 0.34, 0.48	0.10, 0.00, 0.00	0.24, 0.03, 0.05	0.74, 0.09, 0.17	0.76, 0.08, 0.15
Reconnaissance	0.94, 0.94, 0.94	0.56, 0.32, 0.40	0.73, 0.87, 0.79	0.88, 0.92, 0.90	0.90, 0.92, 0.91

Shellcode	0.88, 0.96, 0.92	0.00, 0.00, 0.00	0.71, 0.59, 0.65	0.83, 0.82, 0.83	0.85, 0.83, 0.84
Theft	0.90, 0.97, 0.93	0.00, 0.00, 0.00	0.00, 0.00, 0.00	0.80, 0.96, 0.87	0.85, 0.94, 0.89
Worms	0.85, 0.85, 0.85	0.00, 0.00, 0.00	0.00, 0.00, 0.00	0.33, 0.10, 0.15	0.45, 0.12, 0.19
Injection	0.94, 0.90, 0.92	0.09, 0.00, 0.00	0.67, 0.65, 0.66	0.85, 0.80, 0.82	0.88, 0.82, 0.85
MITM	0.92, 0.40, 0.56	0.00, 0.00, 0.00	1.00, 0.00, 0.01	0.91, 0.24, 0.38	0.90, 0.25, 0.39
Password	0.95, 0.95, 0.95	0.36, 0.92, 0.51	0.65, 0.66, 0.66	0.89, 0.89, 0.89	0.91, 0.91, 0.91
Ransomware	0.99, 0.99, 0.99	0.00, 0.00, 0.00	0.00, 0.00, 0.00	0.95, 0.95, 0.95	0.96, 0.96, 0.96
Scanning	0.99, 0.99, 0.99	0.48, 0.95, 0.64	0.80, 0.79, 0.79	0.97, 0.97, 0.97	0.98, 0.97, 0.97
XSS	0.91, 0.96, 0.93	0.00, 0.00, 0.00	0.62, 0.87, 0.73	0.86, 0.92, 0.89	0.88, 0.94, 0.91

Après balancement :

Mesures	XGBoost	AdaBoost	Logistic Regression	MLP	DNN	LSTM
✓ Accuracy	0.9790	0.4777	0.8752	0.9608	0.9573	0.9650
🔗 Precision	0.9833	0.5482	0.8818	0.9655	0.9622	0.9697
📄 Recall	0.9790	0.4777	0.8752	0.9608	0.9573	0.9650
⚡ F1-Score	0.9798	0.4249	0.8747	0.9613	0.9581	0.9657
🚒 FAR / FPR	0.0010	0.0261	0.0062	0.0020	0.0021	0.0017
📊 ROC-AUC Score	0.9992	0.9631	0.9935	0.9985	0.9985	0.9988
⚙️ MCC	0.9782	0.4711	0.8694	0.9591	0.9554	0.9635

Après PCA : réduction de 44->27

Mesures	XGBoost	AdaBoost	Logistic Regression	MLP	DNN	LSTM
✓ Accuracy	0.9799	0.5859	0.8846	0.9607	0.9670	0.9678
🔗 Precision	0.9840	0.5864	0.8929	0.9647	0.9715	0.9723
📄 Recall	0.9799	0.5859	0.8846	0.9607	0.9670	0.9678
⚡ F1-Score	0.9807	0.5496	0.8848	0.9611	0.9676	0.9684
🚒 FAR / FPR	0.0010	0.0207	0.0058	0.0020	0.0017	0.0015
📊 ROC-AUC Score	0.9992	0.9619	0.9939	0.9984	0.9987	0.9988
⚙️ MCC	0.9791	0.5695	0.8793	0.9589	0.9656	0.9664

Après features importance :

Mesures	XGBoost	AdaBoost
✓ Accuracy	0.9818	0.4902
🎯 Precision	0.9856	0.3974
📄 Recall	0.9818	0.4902
⚡ F1-Score	0.9825	0.4126
🚨 FAR / FPR	0.0009	0.0255
📊 ROC-AUC Score	0.9993	0.9406
⚙️ MCC	0.9810	0.4966

Après PSO :

Meilleur parametres :

Meilleurs paramètres que l'algorithme donne : C=9.62554, max_iter=855

Mesures	Logistic Regression
✓ Accuracy	0.8177
🎯 Precision	0.8219
📄 Recall	0.8177
⚡ F1-Score	0.8138
🚨 FAR / FPR	0.0091
📊 ROC-AUC Score	0.9875
⚙️ MCC	0.8093